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Deployment of Internal Liquid-Liquid Separation Technology to Mitigate Condensate Stabilization Salt Fouling Challenge

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Abstract

This abstract describes the approach taken in a Gas Condensate Stabilization facility to enhance water separation from gas condensate. This was done by installing a liquid-liquid separation technology that was retrofitted inside an existing three-phase separator. This resulted in eliminating a chronic salt fouling challenge in the Condensate Stabilizer Column Reboilers, which was caused by the carryover of the high-salinity water to the Condensate Stabilizer Column due to the inadequate gravity-separation in the existing three-phase separator.

The principle of this proven liquid-liquid separation technology depends on two process stages; a turbulence isolation plate stage that quiescence the liquid flow; and a parallel-plates pack that coalesces small water droplet to form larger droplets which are then separated from the gas condensate by means of gravity. The whole set-up utilizes a self-supporting system that allow for installation inside the existing three-phase separator without any hot work (welding) to the vessel.

The deployment of this technology resulted in achieving high quality gas condensate product, where the free water content dropped by 75%. In addition, it resulted in significant reduction in the capital and operating cost associated with a conventional standalone liquid-liquid coalescer vessel. Furthermore, the enhanced water separation eliminated the chronic salt fouling challenge faced in the Condensate Stabilizer Column Reboilers, thus improving the equipment integrity and availability. Finally, the improved product specifications resulted in smoother operation in the downstream processing facility and pipelines, ensuring a more reliable gas condensate supply network.

Introduction

Liquid-liquid separation is a critical process in the oil and gas industry, particularly in gas condensate stabilization systems where the efficient removal of free water from condensate is essential to meet product specifications and prevent operational issues such as salt fouling in reboilers. This manuscript presents the results of deploying a high-efficiency internal technology designed to enhance liquid-liquid separation in a gas condensate stabilization system. The test was carried out at a gas processing facility experiencing operational challenges related to slugging and salt fouling.

The internal liquid-liquid separation technology was installed in an existing condensate flash drum as part of a pilot project to assess its performance under field conditions. The test was conducted over a 20-day period, during which the condensate feed flowrate was incrementally increased to the design flowrate. The results of the test, including the removal efficiency of free water and the impact on reboiler fouling, are discussed in detail.

Background

Fig. 1 below represents a typical setup for a Gas Condensate Stabilization facility. Normally, the gas feed, which typically consist of 3 phases (gas, condensate and water), is fed into a Slug Catcher. The Slug Catcher is essentially a 3-phase separator that removes the condensate and water from the gas capitalizing on the density variance between the 3 phases. The condensate is further separated in the Condensate Flash Drum, where flashed gas and sour water are removed from the condensate. The condensate is then processed at the Stabilizer Column, equipped with reboilers to heat-up the condensate in order to remove any entrained light-end hydrocarbon (i.e. gas) from the condensate, thus stabilizing the condensate to meet the product specifications.

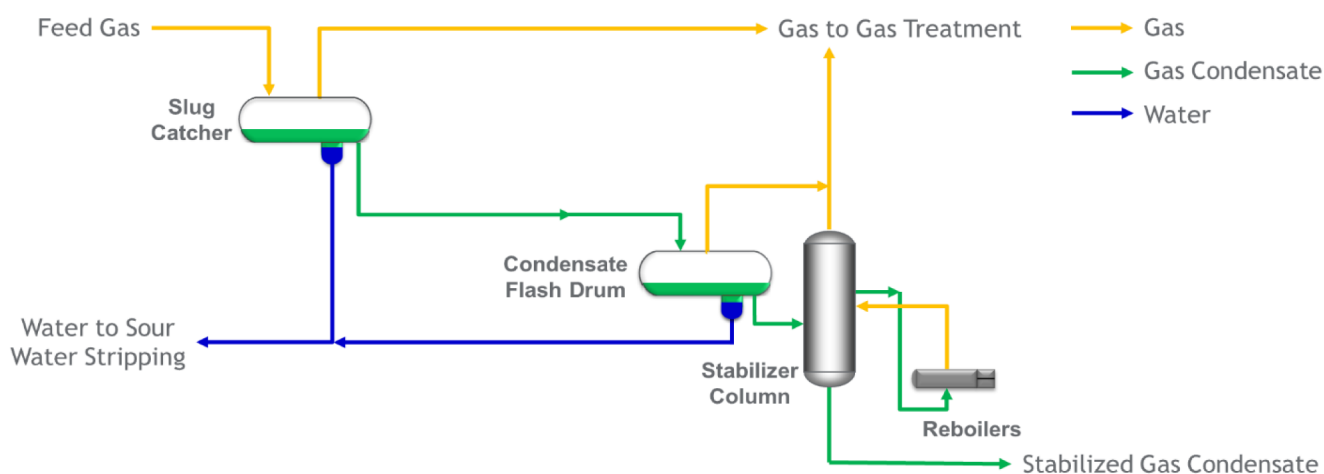


Figure 1—typical setup of a gas condensate stabilization facility

The main limitation in the above setup is the reliance on gravity separation for water-condensate separation. While gravity separation is effective for removing bulk water from condensate, it is not as effective when it comes to removing the free water droplets suspended in the condensate phase.

As a result of this limitation, one of the common challenges faced in such Gas Condensate Stabilization facilities is the carryover of the free water droplets into the Stabilizer Column and from there into the Reboilers. Considering the high operating temperature inside the Reboilers, these water droplets evaporate, leaving the dissolved salt to deposit inside the Reboilers as illustrated by Fig. 2 below. As this process continues, salt fouling accumulates around the Reboilers tubes, which reduces the heating efficiency of the Reboilers and eventually reaches to a point where the Reboilers can no longer achieve the desired temperature. This mandate shutting down the unit to conduct mechanical cleaning in order to restore the performance of the Reboilers and maintain the gas condensate product specifications.

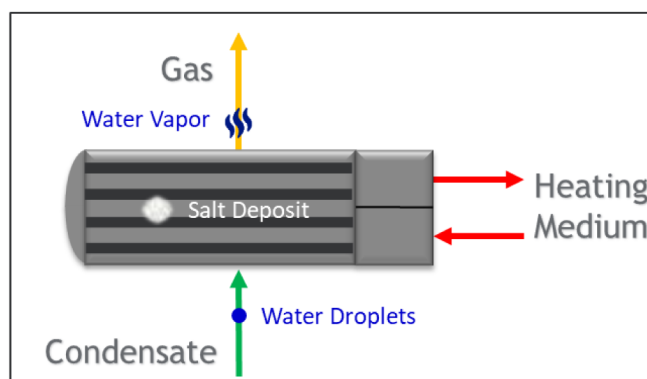


Figure 2—formation of salt fouling in the gas condensate stabilizer reboiler due to water droplet carryover

Objective

A common way to address the salt fouling challenge in the gas condensate stabilization facility is to install a Liquid-Liquid Coalescer (LLC) between the condensate flash drum and stabilizer column. An LLC is a device designed to enhance the separation of two immiscible liquids, such as gas condensate and water, by facilitating the merging of tiny droplets into larger ones. This process, known as coalescence, occurs as the mixture passes through a medium, such as fibers or plates, which encourages droplets to combine, forming bigger and heavier drops that settle more rapidly. Unlike conventional gravity-based separation, where separation relies solely on the natural settling of denser liquids, coalescers actively promote droplet merging, leading to faster and more efficient separation. This makes coalescers particularly effective in handling higher flow rates and mixtures with similar densities where gravity alone is insufficient, such as the case in the Condensate Stabilization system.

One of the main drawbacks of LLC is the high capital cost involved. Typically, two separate vessels are required for conventional LLC. The first vessel is a pre-filtration system designed to remove suspended solid particles. The second vessel is where the actual liquid-liquid coalescing process occurs. Consequently, the conventional LLC technology requires large construction, piping and instrumentation work, which significantly boost the capital cost associated with installing this technology.

In order to optimize the capital cost required to address the condensate stabilization salt fouling challenge, an alternative Internal Liquid-Liquid Separation (ILLS) technology was considered to be deployed. This technology works in similar manner to conventional LLC, with the exception that the coalescing materials (i.e. the plates) are installed inside an existing drum, in this particular case the condensate flash drum. By retrofitting the liquid-liquid coalescer inside an existing vessel, the construction, piping and instrumentation work is eliminated, resulting in significantly reducing the capital cost.

The objective of this paper is to explore the key differences between conventional LLC and the ILLS technology. In addition, a performance test for the ILLS will be presented for a gas processing facility that faced a chronic salt fouling challenge at its gas condensate stabilization facility. The result and future work for this technology will also be discussed in details.

Technologies Overview

Conventional Liquid-Liquid Coalescer (LLC)

Typical liquid-liquid coalescing systems in condensate stabilization facilities utilize a two-stage approach as shown in Fig. 3 below. The first stage is a pre-filter vessel designed to remove suspended solid particles. The second stage is a Liquid-Liquid Coalescer vessel which utilizes high performance cartridge filter elements to remove discontinuous phase from continuous as shown in the below figure.

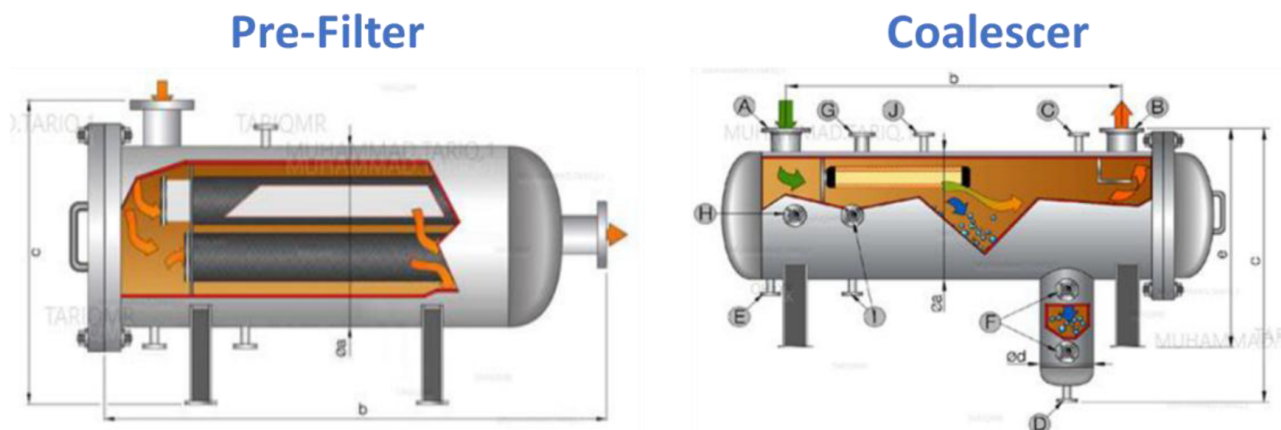


Figure 3—the two components of a conventional liquid-liquid coalescing system; a pre-filter and a coalescer

The system consists of perforated basket assemblies attached to an element support plate at the inlet and supported by a lattice grid at the outlet. These baskets are divided into three parts: the perforated basket, cartridge sleeve, and rear support post or bayonet. The sleeve is welded to the support plate, and cartridges slide into the baskets, sealed by an ‘O’ ring to ensure no leaks.

At the outlet end, the cartridges are capped for easy removal, facilitating maintenance. The system also includes an element support plate with flared riser pipes and a z-bars assembly that provides strong support for the cartridges. The cartridges are sealed at both ends, with the outlet end using a dimple washer, stud, and double nuts for secure fastening. Additionally, a baffle directs contaminant droplets downward, enhancing the filtration process.

Internal Liquid-Liquid Separation (ILLS)

The new high efficiency internals technology consists of a new coalescing section with Parallel Plate Liquid-Liquid Coalescing Media and Self-Supporting Expansion Rings to be mounted and installed inside the existing Flash Drum as shown in Fig. 4 below.



Figure 4—the components of internal liquid-liquid separation technology

The set-up consists of two sections. The first section contains a Turbulence Isolation Plate to be installed to quiescence the liquid flow prior to entering Section 2. The second section is where the liquid-liquid separation takes place, where a Plate Pack Coalescer Media is installed to separate the free water from the condensate up to the desired target. The Self-Supporting Expansion Rings system allow for installation with no welds to the existing vessel wall.

Technologies Comparison

Removal Efficiency

At the process conditions in the facility where the salt fouling challenge was encountered, simulations show that the LLC can reduce the free water content in the gas condensate from 5,000 ppm down to <20 ppm. On the other hand, the ILLS Technology was simulated to achieve 15.8 ppm free water content at similar conditions. This indicates that both technologies are expected to produce comparable results in terms of removal efficiency.

Mechanical Work

The conventional LLC requires the installation of two new vessels, namely the pre-filter and the liquid-liquid coalescing vessel. In addition, new associated piping, valves and instrumentation need to be installed. This makes the LLC a mechanically demanding project that involves a lot of construction and hot work.

On the other hand, the ILLS is retrofitted completely inside the existing three-phase separator without the need to install any new equipment. Furthermore, thanks to the Self-Supporting Expansion Rings system, no welding is required to install the ILLS inside the existing vessel. This significantly optimizes the construction and mechanical work involved to deploy this technology, especially when compared to the conventional one.

Operability

The installation of a new pre-filtration system for the conventional LLC introduces new consumables that require frequent maintenance and operation intervention. The filter elements will need to be replaced on a frequent basis to maintain the differential pressure across the filters and hence sustain the gas condensate flow into the LLC.

In contrast, the ILLS Technology does not entail any new consumables. In fact, the only operating cost associated with the technology is to conduct periodic cleaning at the already-scheduled Turnaround & Inspection for the condensate stabilization system. Hence, there is minimal operating cost associated with the installation of this technology.

Financial Cost

Considering the difference in the mechanical scope between the technologies, it is clear that the ILLS Technology has a lower capital cost. In fact, it was estimated that the cost of this technology is only 9% of the cost of a Conventional LLC. Moreover, there is minimal operating cost associated with the new technology, while the conventional one will entail operating cost for the periodic filter elements replacement.

Performance Test

Test Preparation

The performance test was conducted in the facility where the salt fouling challenge was encountered over a 20-day period. The ILLS technology was installed inside the condensate flash drum of the Gas Condensate Stabilization facility, and the test was conducted at incremental condensate feed flowrates, ranging from 70% to 100% of the facility's capacity.

At each test point, inlet and outlet condensate samples were collected for analysis of free water content using the Basic Sediment & Water (BS&W) laboratory method. The BS&W method is a standard technique for determining the water content in hydrocarbon liquids and is widely used in the oil and gas industry for this purpose.

Operating Parameters

The operating parameters during the test were carefully monitored to ensure that the conditions were representative of normal operations. The key parameters included:

Condensate feed flowrate. the flowrate was incremented in steps from 70% to 100% of the stabilizer facility's capacity to evaluate the performance of the technology under varying loads as shown in Fig. 5.

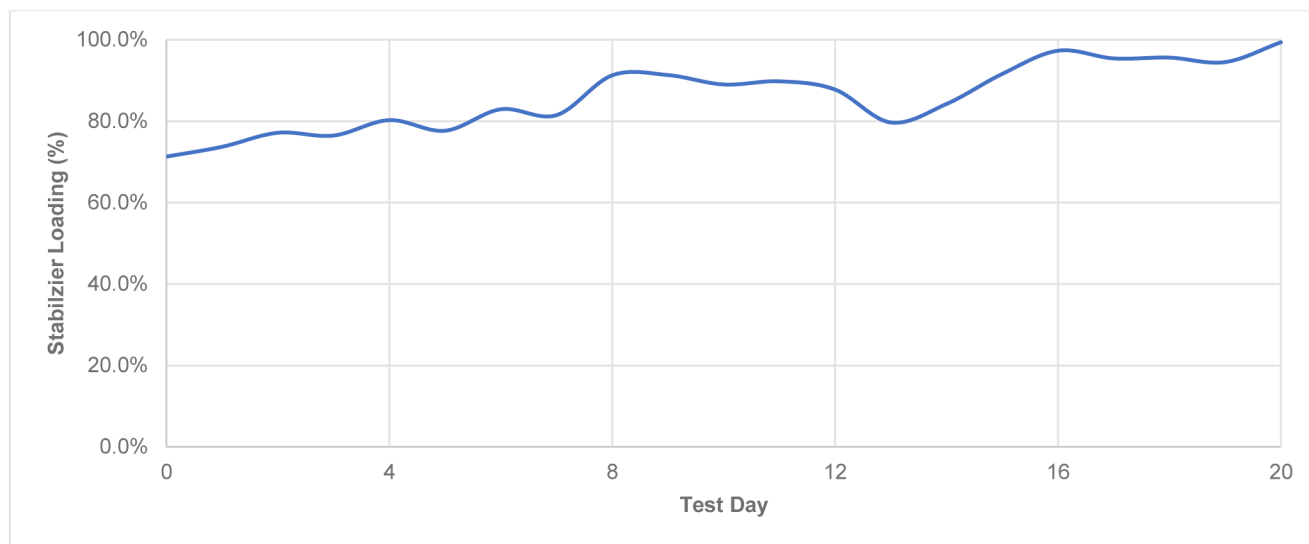


Figure 5—the condensate feed rate during the test period as a % of the total stabilizer unit capacity

Condensate flash drum liquid levels. the water and gas condensate level inside the vessel where ILLS Technology was installed were monitored and confirmed to be stable at the normal level. This is illustrated by Fig. 6 below. The only exception was a slight increase in the water level on day 2 of the test due a control valve tuning issue that was addressed immediately. Similarly, the respective level control valves for the two phases were confirmed to be maintained at normal % opening well below 100%, indicating that the amount of liquid and the water content in the Gas Condensate feed were maintained within the capacity of the level control valves.

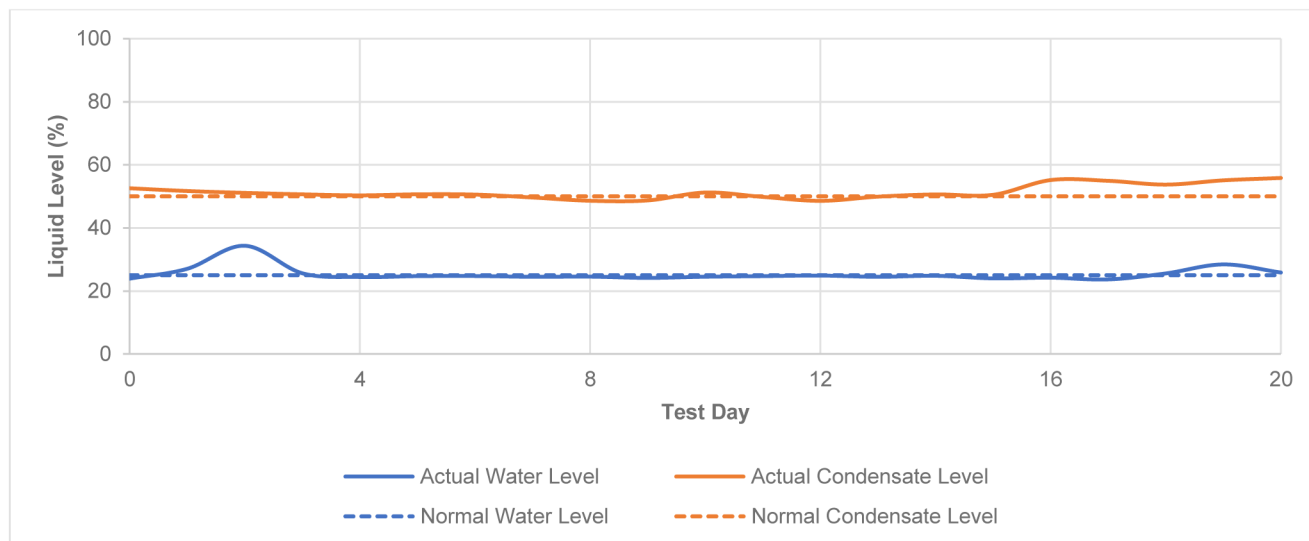


Figure 6—water and condensate level in the condensate flash drum

Condensate flash drum pressure and temperature. the ILLS technology is not expected to cause any significant variation in the pressure and temperature profile inside the vessel where it is installed. The only exception is a minor pressure drop across the coalescing medium. This was evident during the test, where the temperature was well maintained within the limit. Additionally, there was no noticeable variation in terms of pressure.

Results and Analysis

Lab Results

Several samples were collected during the test to assess the performance of the technology. Samples were taken from both upstream and downstream the Condensate Flash Drum, with an objective to assess the removal efficiency of the technology by measuring the free water content through the BS&W method. Furthermore, samples were analyzed at different laboratories (Lab 1 and Lab 2). The objective of this is to ensure that the lab results are not biased due to sample analysis error associated with the apparatus used in each laboratory.

Table 1 below summarizes the lab results:

Table 1—summary of lab analysis results

Laboratory	Stabilizer Loading (%)	Condensate Feed BS&W (Vol %)	Condensate Product BS&W (Vol %)	Water Removal Efficiency (%)
Lab 1	71%	1.27	0.3	76.4%
Lab 1	71%	0.9	0.4	55.6%
Lab 1	78%	1.1	0.6	45.5%
Lab 1	78%	0.14	0.06	57.1%
Lab 1	78%	0.12	0.04	66.7%
Lab 1	78%	0.12	0.04	66.7%
Lab 1	90%	0.06	0.03	50.0%
Lab 1	90%	0.1	0.04	60.0%
Lab 1	90%	1.8	0.04	97.8%
Lab 1	100%	2.3	1	56.5%
Lab 1	83%	1.3	0.7	46.2%
Lab 2	71%	0.1	0.05	50.0%
Lab 2	71%	0.1	0.05	50.0%
Lab 2	78%	0.1	0.05	50.0%
Lab 2	78%	0.1	0.025	75.0%
Lab 2	90%	0.05	0.025	50.0%
Lab 2	90%	0.1	0.05	50.0%
Lab 2	83%	0.05	0.025	50.0%
Lab 2	97%	0.35	0.1	71.4%
Lab 2	87%	0.4	0.15	62.5%
Lab 2	90%	0.1	0.05	50.0%
Lab 2	90%	0.25	0.1	60.0%
Lab 2	100%	0.62	0.3	51.6%

There are two main independent variables in assessing the performance of the ILLS, which are the Stabilizer Loading and the Condensate Feed BS&W.

Stabilizer Loading Impact Analysis

The flowrate to the unit (or the Stabilizer Loading, taken as the actual condensate feed flowrate divided by the Stabilizer unit capacity) is a key independent variable that is expected to impact the performance of the Stabilizer unit. Generally, liquid-liquid separation favors lower flowrate as this would allow higher residence time. Additionally, lower flowrate is more likely to maintain laminar flow, which improves the coalescing effect.

From Fig. 7, there does not seem to be a strong correlation between the Stabilizer Loading and Water Removal Efficiency. However, Fig. 8 demonstrates a clearer relationship between the Stabilizer Loading and the Condensate Product BS&W, where higher condensate flowrate leads to more BS&W content in the product. This is aligned with the expected impact of higher flowrate.

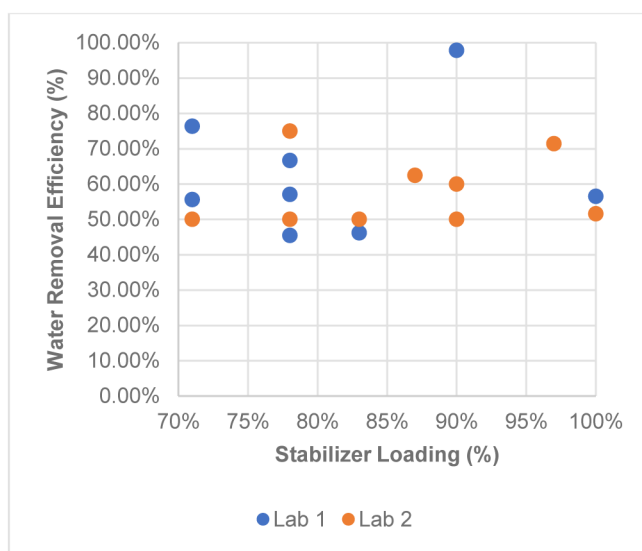


Figure 7—impact of stabilizer loading on water removal efficiency

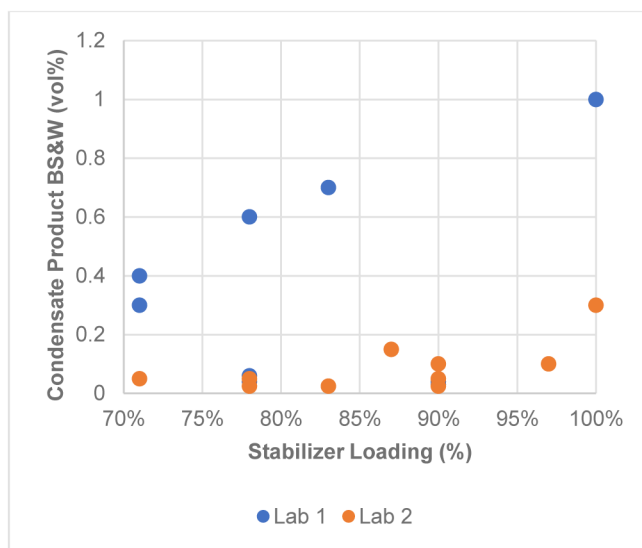


Figure 8—impact of stabilizer loading on condensate product BS&W

One observation from Fig. 7 and Fig. 8 below is that Lab 2 does not seem to be able to capture higher amount of BS&W content above 0.4%, which may indicate limitation in this laboratory apparatus. This may suggest that Lab 1 provides more reliable data to assess the performance of the technology.

Condensate Feed BS&W Impact Analysis

When higher water content is present in the condensate feed, the LLS Technology might be overloaded by the excess water, which would lead to lower performance. This anticipated impact was not clearly seen in Fig. 9, when water removal efficiency was plotted against the condensate feed BS&W. However, the correlation with the condensate product BS&W is much clearer as shown in Fig. 10, where almost linear relationship is demonstrated between the two variables. These trends might indicate that using the condensate product BS&W as the dependent variable gives better representation into the performance of the technology.

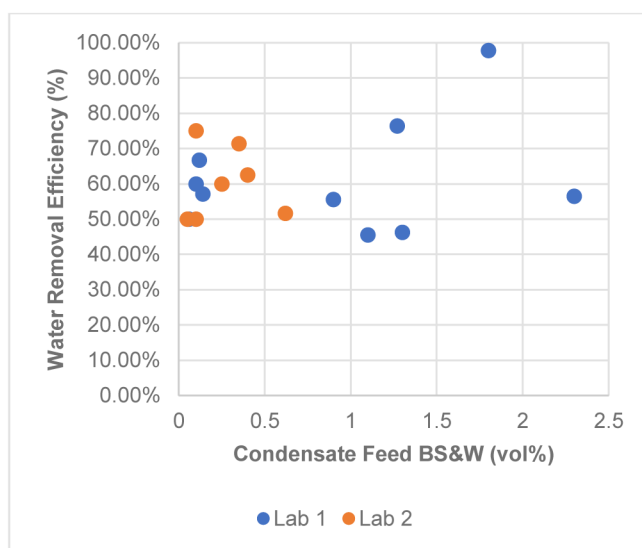


Figure 9—impact of condensate feed BS&W on water removal efficiency

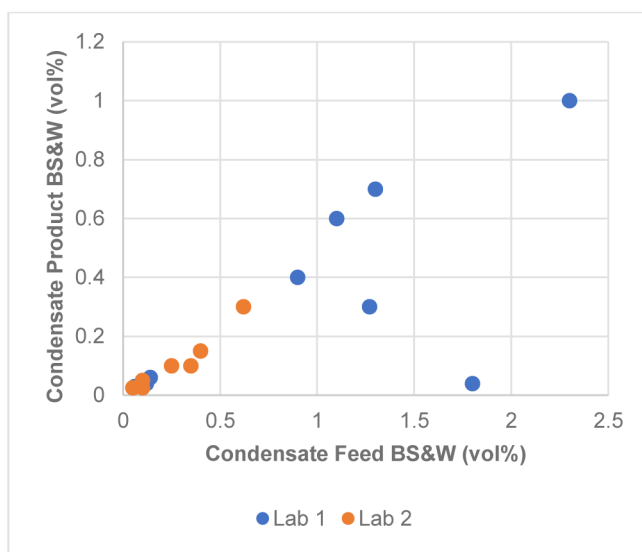


Figure 10—impact of condensate feed BS&W on condensate product BS&W

Furthermore, Fig. 10 indicates once more the limitation in Lab 1 BS&W analysis, where the highest recorded BS&W value is 0.6%. This point is noted during the performance test, but is not the focus on this manuscript.

Overall Performance

The performance test demonstrated that the high-efficiency internals technology is capable of significantly reducing the free water content in the condensate feed from more than 2% BS&W to less than 1%, achieving an average removal efficiency of 75%. The technology also succeeds in maintaining the condensate product specification within the desired value, but fails to achieve the technology design specification of reducing the BS&W content to less than 20 ppm (0.002%).

Future Work

The most likely cause behind the lower than expected performance of the technology is the water droplet size. Although water droplets were not measured during the design or test phase of the LLS Technology, the design basis assumed an average water droplet size of 120 micron. However, considering the performance test results and through reverse engineering, it was estimated that the average water droplet size is actually 28 micron. Smaller water droplets are much harder to merge together by the coalescing effect, hence why the performance of the technology is strongly impacted by this key variable.

To further improve the technology performance, it is proposed to install a second layer of plate packs with a different configuration that have tighter spacing between the layers as shown in Fig. 11. The new proposed design is expected to improve the separation efficiency as it will result in less distance for the coalesced droplets to fall along with the reduced turbulence.

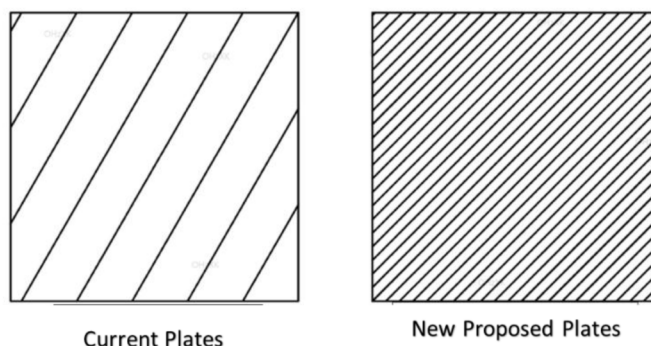


Figure 11—current and new proposed LLS plate packs

Conclusion

The Internal Liquid-Liquid Separation Technology proved to be an effective solution for mitigating salt fouling challenges in the Gas Condensate Stabilization process. By achieving a significant reduction in free water content—from 2000 ppm to less than 1000 ppm—the technology eliminated the chronic fouling issues in the reboilers, thereby enhancing equipment integrity and operational reliability. The retrofitting of this technology within the existing three-phase separator not only avoided the need for costly standalone vessels but also minimized installation complexity by eliminating hot work. The improved water separation also ensured higher-quality condensate, leading to smoother operations in downstream facilities and pipelines. This case study underscores the value of advanced separation technologies in addressing operational challenges while reducing costs and improving overall system performance.

The performance of the technology can be further improved with the introduction of a second plate pack layer with a tighter plate spacing. The new configuration is expected to achieve 99.9% separation efficiency for water droplet of size 35 micron and larger, compared to around 75% for the current configuration.